1. Newsvendor

- Demand is random
- Single period

How much should be produced every day?

Too much v. Too little

Demand uncertainty:

- Demande is a **random variable**
- We don't know the **realization**

$$D \sim N(\mu, \sigma)$$

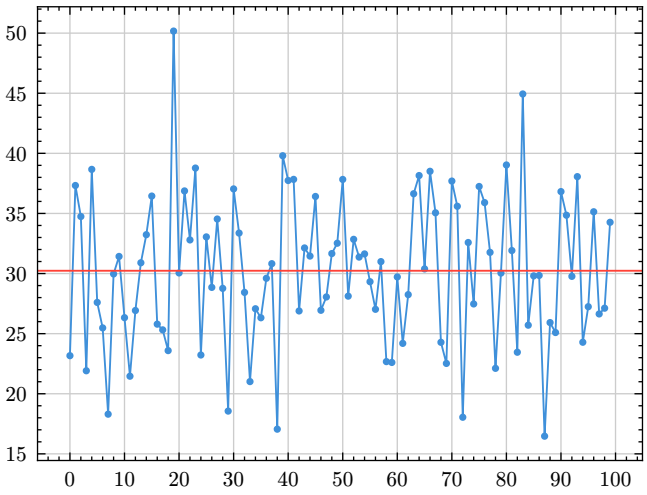
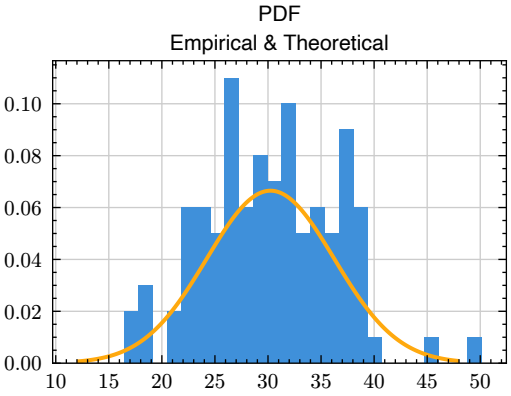
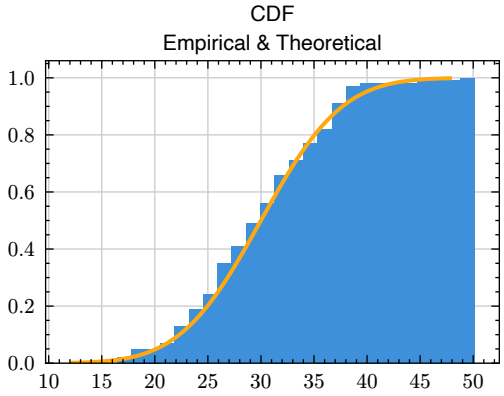
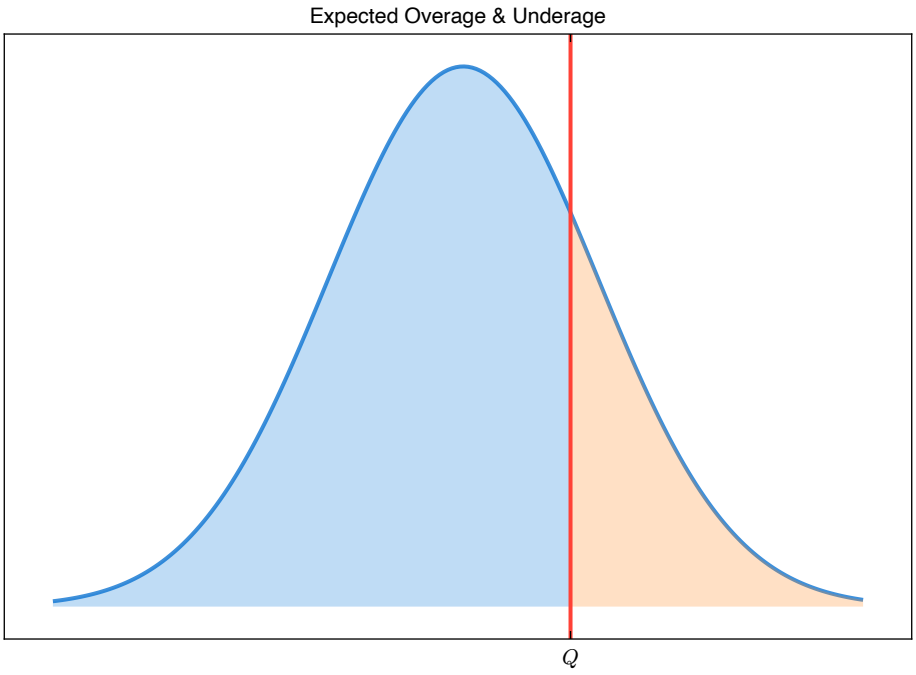


Figure 1: Random Demand





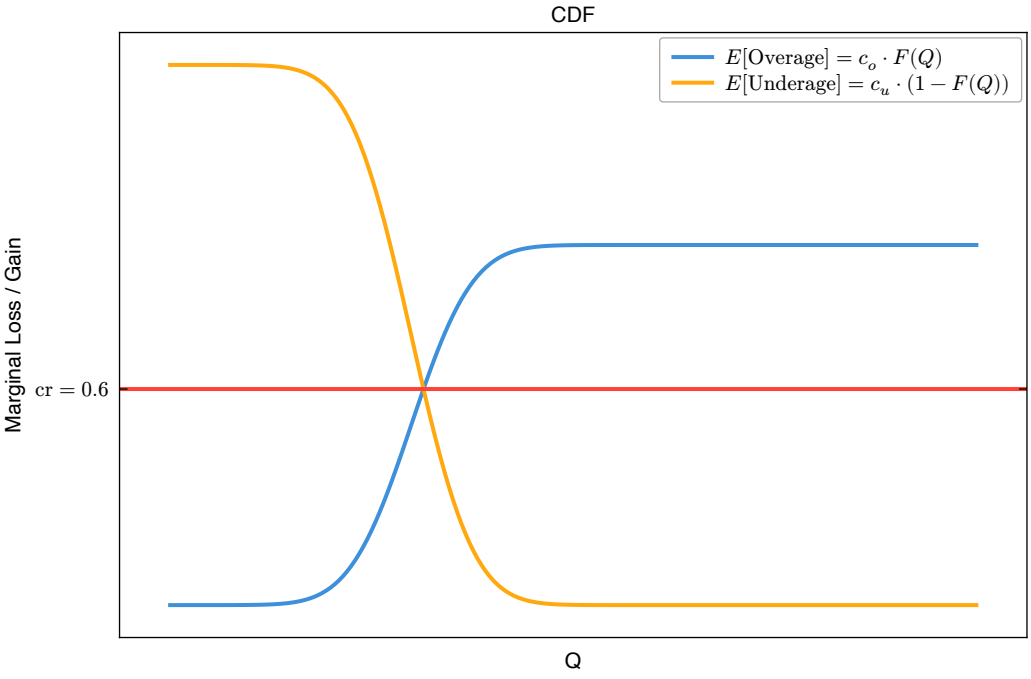


Expected overage (marginal)

$$c_o \cdot F(Q)$$

Expected underage (marginal)

$$c_u \cdot (1 - F(Q))$$



Optimal production quantity:

$$c_o \cdot F(Q^*) = c_u \cdot (1 - F(Q^*))$$
$$F(Q^*) = \frac{c_u}{c_u + c_o}$$

Type I service level (a): probability of **not** stocking out.

The cost-optimal Q^* corresponds to $a = \frac{c_u}{c_u + c_o}$.

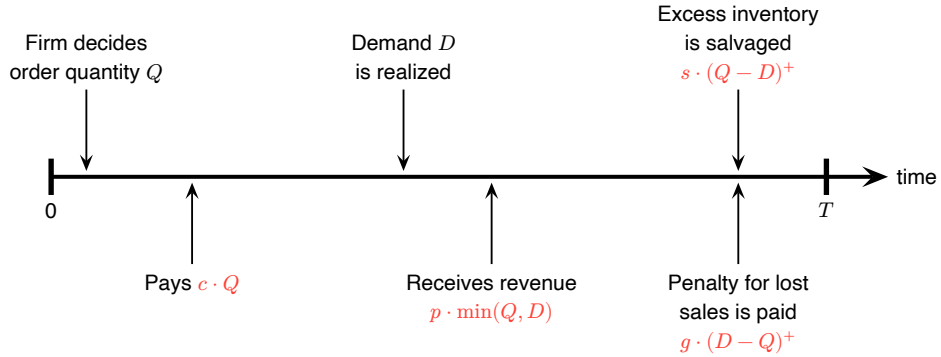
$$F(Q^*) = a$$
$$Q^* = F^{-1}(a) = \mu + z(a)\sigma \quad \text{if } D \sim N(\mu, \sigma^2)$$

where $z(a) = \Phi^{-1}(a)$ is the inverse of the standard normal CDF.

```python
from scipy.stats import norm

a = norm.cdf(Q, loc=mu, scale=sigma)    # Q -> service level
Q = mu + norm.ppf(a) * sigma            # service level -> Q
Q = norm.ppf(a, loc=mu, scale=sigma)    # equivalently
```

- D: demand
- Q: order quantity
- p: unit revenue
- c: unit cost
- s: salvage vaue per unit leftover
- g: goodwill cost per unit short



Objective function

$$\pi(Q) = E[\underbrace{p \cdot \min(Q, D)}_{\text{sold}} + \underbrace{s \cdot (Q-D)^+}_{\text{salvage}} - \underbrace{g \cdot (D-Q)^+}_{\text{short}} - \underbrace{c \cdot Q}_{\text{purchasing}}]$$

Rewrite:

$$\min(D, Q) = D - (D-Q)^+$$

- $D > Q \to Q$
- $D < Q \to D$

$$Q = D - (D-Q)^+ + (Q-D)^+$$

- $D > Q \to Q$
- $D < Q \to Q$

We get:

$$\pi(Q) = \underbrace{(p-c)E[D]}_{\substack{\text{Profit} \\ \text{(Perfect Demand} \\ \text{Information)}}} - E[\underbrace{(p+g-c) \cdot (D-Q)^+}_{\text{Underage}} + \underbrace{(c-s) \cdot (Q-D)^+}_{\text{Overage}}]$$

The profit maximization problem is equivalent to minimizing of:

$$C(Q) = E[c_u \cdot (D-Q)^+ + c_o \cdot (Q-D)^+]$$

Where:

- $c_u = p + g - c$
- $c_o = c - s$

$$\frac{\mathrm{d}C(Q)}{\mathrm{d}Q} = c_o F(Q) - c_u(1 - F(Q)) = 0$$

$$F(Q^*) = P(D \le Q^*) = \frac{c_u}{c_u + c_o} = \frac{p - c + g}{p - s + g} = \alpha$$

Minimize the expected total cost of underage and overage, choose Q such that the probability that we don't have lost sales (i.e., demand is Q or lower) equals the critical ratio.

Example: O'Neill's Hammer 3/2 Wetsuit

- Demand mean (μ_D): 3192 units / period
- Demand std (σ_D): 1181 units / period

- Price (p): 190 $ / unit sold
- Cost (c): 110 $ / unit bought
- Salvage (s): 90 $ / unit leftover
- Goodwill (g): 0 $ / unit short

$$\text{cr} = \frac{80}{80+20} = 0.8$$

$$z = 0.84$$

$$\begin{aligned} Q &= \mu + z(\alpha) \cdot \sigma \\ &= 3192 + 0.84 \cdot 1181 \\ &= 4185.95 \end{aligned}$$

1.1. Performance Measures

1.1.1. In-stock Probability (Type I - Service Level)

Probability all demand is satisfied

$$D \leq Q$$

$$F(Q) = \Phi(z)$$

```
from scipy.stats import norm

z = (Q - mu) / sigma
instock = norm.cdf(z)
```

Choose Q subject to a minimum in-stock probability

```
from scipy.stats import norm

p_instock = 0.99

z = norm.ppf(p_instock)
Q = mu + z * sigma
Q = norm.ppf(p_instock, loc=mu, scale=sigma)
```

1.1.2. Stockout Probability

$$P(\text{Stockout}) = P(D > Q) = 1 - F(Q) = 1 - \Phi(z), \quad z = \frac{Q-\mu}{\sigma}$$

The complement of the Type I service level: $P(\text{Stockout}) = 1 - a$. At the cost-optimal Q^*:

$$P(\text{Stockout}) = 1 - \frac{c_u}{c_u + c_o} = \frac{c_o}{c_u + c_o}$$

```
from scipy.stats import norm

z = (Q - mu) / sigma
p_stockout = norm.sf(z)
p_stockout = norm.sf(Q, loc=mu, scale=sigma)
```

1.1.3. Fill Rate (Type II - Service Level)

$$E[\text{Fillrate}] = \frac{E[\text{Sales}]}{\mu} = 1 - \frac{E[\text{Short}]}{\mu}$$

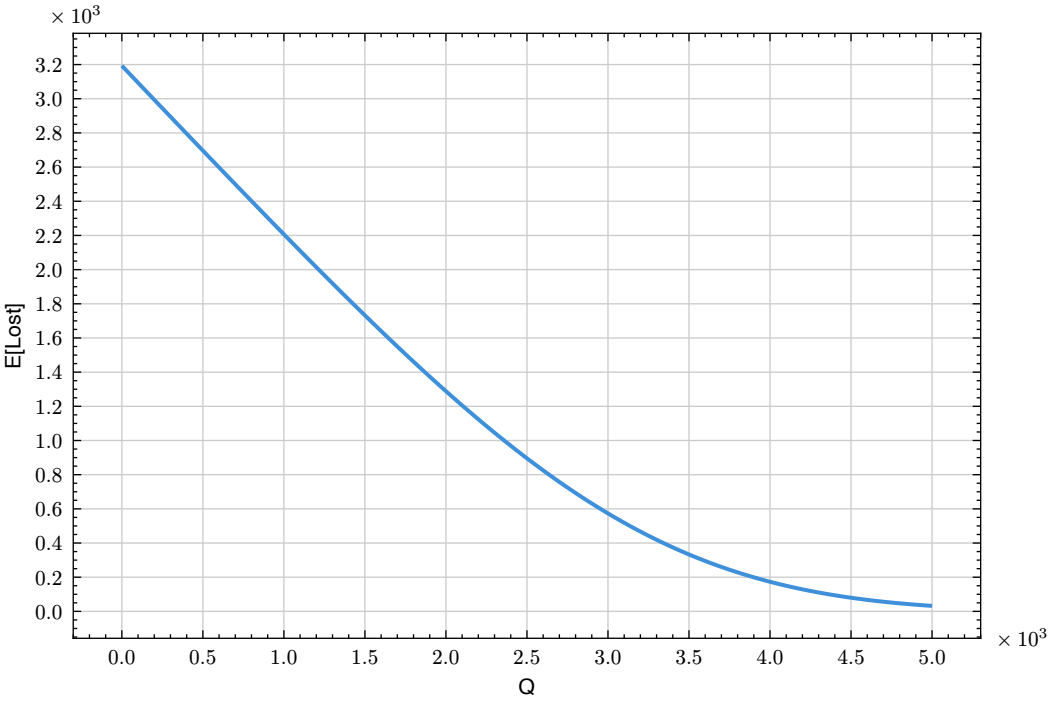
1.1.4. Expected Lost Sales (Shortage)

$$E[\text{Short}] = E[(D - Q^+)] = \int_Q^\infty (D-Q) f(D)\, \mathrm{d}D$$

$$L(z) = \varphi(z) - z(1 - \Phi(z))$$

```
from scipy.stats import norm

z = (Q - mu) / sigma
L = norm.pdf(z) - z * (1 - norm.cdf(z))
e_short = sigma * L
e_short = sigma * (norm.pdf(z) - z * norm.sf(z))
```



1.1.5. Expected Sales

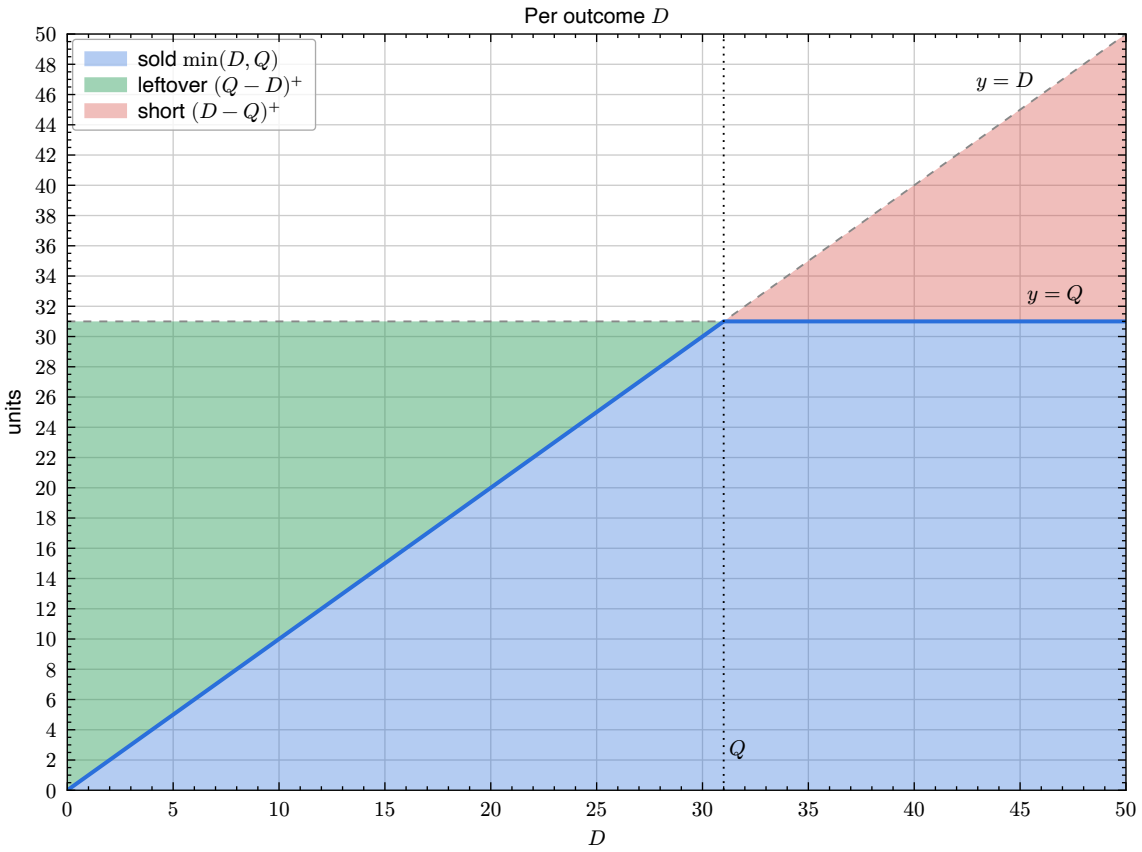
$$E[\text{Sold}] = E[\min(D, Q)] = E[D - (D - Q)^+] = \mu - E[\text{Short}]$$

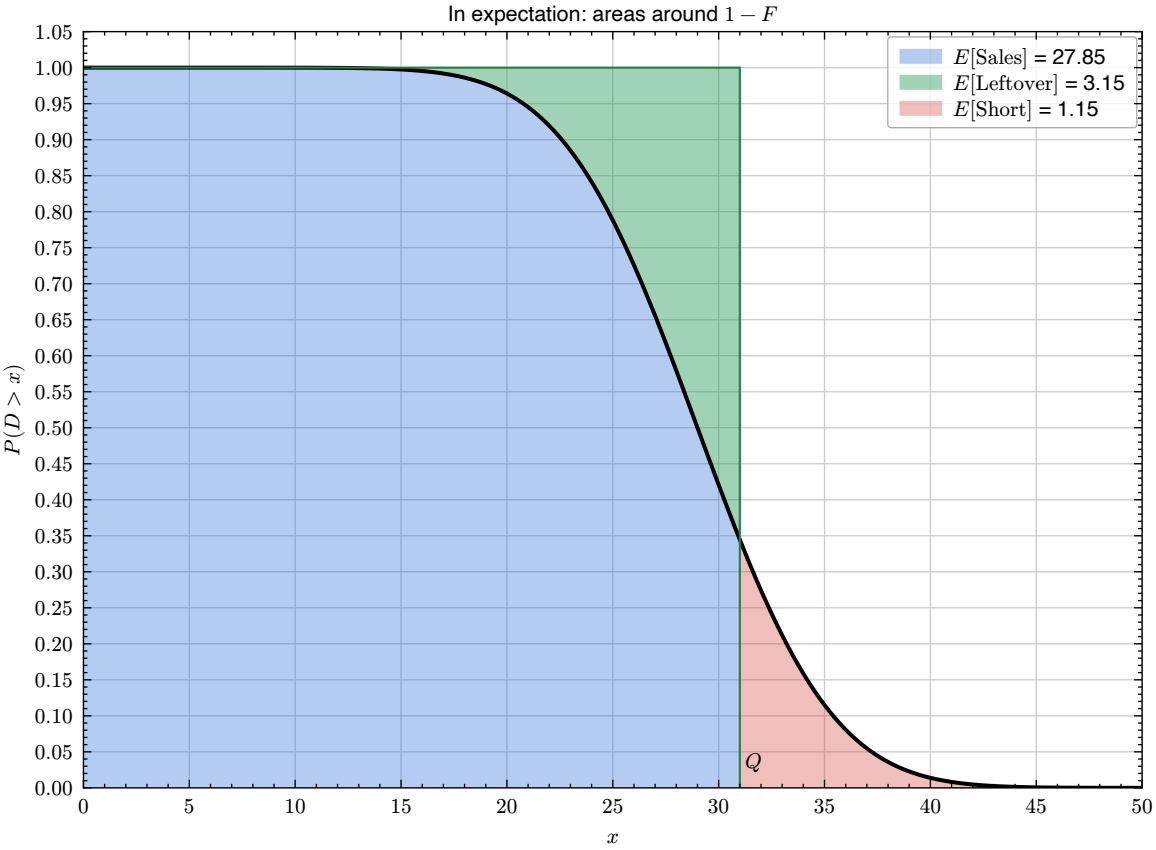
```
z = (Q - mu) / sigma
e_short = sigma * (norm.pdf(z) - z * norm.sf(z))
e_sales = mu - e_short
```

1.1.6. Expected Leftover

$$E[\text{Leftover}] = Q - E[\text{Sales}] = Q - \mu + E[\text{Short}]$$

```
z = (Q - mu) / sigma
e_short = sigma * (norm.pdf(z) - z * norm.sf(z))
e_sales = mu - e_short
e_leftover Q - e_sales
```





blue + red	=	total area under $1 - F = \mu$	$\Rightarrow E[\text{Sales}] = \mu - E[\text{Short}]$
blue + green	=	rectangle $Q \times 1 = Q$	$\Rightarrow E[\text{Leftover}] = Q - E[\text{Sales}] = Q - \mu + E[\text{Short}]$

1.1.7. Expected Profit

$$E[\text{Profit}] = \pi(Q) = pE[\text{Sales}] + sE[\text{Leftover}] - gE[\text{Short}] - cQ$$

Example

Expected profit if we order 4196 units of Hammer 3/2s

- Order quantity (Q): 4196 units
- Demand mean (μ_D): 3192 units / period
- Demand std (σ_D): 1181 units / period
- Price (p): 190 $ / unit sold
- Cost (c): 110 $ / unit bought
- Salvage (s): 90 $ / unit leftover
- Goodwill (g): 0 $ / unit short

$$z = \frac{Q - \mu}{\sigma} = \frac{4196 - 3192}{1181} = 0.85$$

$$L(z) = \varphi(z) - z(1 - \Phi(z)) = 0.11$$

$$E[\text{Short}] = \sigma L = 129.85$$

$$E[\text{Sales}] = \mu - E[\text{Short}] = 3062.15$$

$$E[\text{Leftover}] = Q - E[\text{Sales}] = 1133.85$$

$$\begin{aligned} E[\text{Profit}] &= pE[\text{Sales}] + sE[\text{Leftover}] - gE[\text{Short}] - cQ \\ &= 222295.3 \end{aligned}$$

$$E[\text{Fill}] = \frac{E[\text{Sales}]}{\mu} = 0.96$$

$$E[\text{InStock}] = \Phi(z) = 0.8$$

$$E[\text{Stockout}] = 1 - \Phi(z) = 0.2$$

1.2. Poisson Distribution

PMF

$$P(D = x) = \frac{\lambda^x e^{-\lambda}}{x!}$$

CDF

$$F(Q) = P(D \le Q) = \sum_{x=0}^{Q} P(x) = \sum_{x=0}^{Q} \frac{\lambda^x e^{-\lambda}}{x!}$$

$$E[\text{Lost}] = \lambda - \lambda F(Q-1) - Q(1 - F(Q))$$

Example: Tea World

6

0.32311651547875186

1.3. Double Marginalization: Buyback Contract

What's given	What you do	Formula
w	Solve for b	$b = p + r - (p - w) \cdot \frac{p-s}{p-c}$
b	Solve for w	$w = p - (p - b + r) \cdot \frac{p-c}{p-s}$
Neither: "for which w does it work?"	Give the feasible range	$c < w < p - r\frac{p-c}{c-s}$

What's given	What you do	Formula
Neither: "which w would both accept?"	Give the win-win range	$w_0 + \frac{\Pi_S^0 - \Pi_S(w_0)}{m} \le w \le w_0 + \frac{\Pi_R(w_0) - \Pi_R^0}{m}$ with $m = Q^* - L\frac{p-s}{p-c}$

E[Shortage]	$\varphi(z) - z \cdot (1i\Phi(z))$
E[Leftovers]	$z \cdot \Phi(z) + \varphi(z)$

Step 1. Find the best order quantity for the whole chain

- $c_u = p - c = 150, \quad c_o = c - s = 25$
- $\mathrm{CR} = \frac{c_u}{c_u + c_o} = 0.857$
- $z = \Phi^{-1}(\mathrm{CR}) = 1.068$
- $Q^* = \mu + z\sigma = 1320$

Step 2. Find the buyback price b for the given w

Choose b so that SkiPro's critical ratio equals the chain's:

$$\frac{p-w}{p-b+r} = \frac{p-c}{p-s} \quad \Rightarrow \quad b = p + r - (p-w)\frac{p-s}{p-c} = 167.5$$

With this b, SkiPro's own critical ratio is 0.857, so it orders $Q^* \approx 1320$.

Step 3. Check the conditions

- $s + r < b < w$: $80 < 167.5 < 175$ ✓
- Holds for any w with $c < w < p - r\frac{p-c}{c-s}$, i.e. $100 < w < 220$

Step 4. Calculate each firm's expected profit with the buyback

- $I = E[\text{leftovers}] = \sigma[z\Phi(z) + \varphi(z)] = 342$
- $S = E[\text{sales}] = Q^* - I = 978$

	Formula	Value
SkiPro	$\Pi_R^{\mathrm{BB}} = pS + (b-r)I - wQ^*$	69077
Obermeyer	$\Pi_S^{\mathrm{BB}} = (w-c)Q^* - (b-s)I$	67366
Total	$\Pi_R^{\mathrm{BB}} + \Pi_S^{\mathrm{BB}}$	136443

Step 5. Calculate each firm's profit without the buyback

Wholesale-only contract: SkiPro salvages its own leftovers.

- $c_u = p - w = 75, \quad c_o = w - s = 100$
- $\mathrm{CR}^W = \frac{c_u}{c_u + c_o} = 0.429$
- $z^W = \Phi^{-1}(\mathrm{CR}^W) = -0.18$
- $Q^W = \mu + z^W\sigma = 946$
- $I^W = \sigma[z^W\Phi(z^W) + \varphi(z^W)] = 95$
- $S^W = Q^W - I^W = 851$

	Formula	Value
SkiPro	$\Pi_R^W = pS^W + sI^W - wQ^W$	54392
Obermeyer	$\Pi_S^W = (w-c)Q^W$	70950
Total	$\Pi_R^W + \Pi_S^W$	125342

Step 6. Compare the two contracts

	Wholesale	Buyback	Change
SkiPro	54392	69077	14685

	Wholesale	Buyback	Change
Obermeyer	70950	67366	–3584
Total	125342	136443	11101

The total goes up, but at $w = 175$ Obermeyer is worse off than without the buyback, so it wouldn't offer this deal.

Step 7. Find the win-win range for w

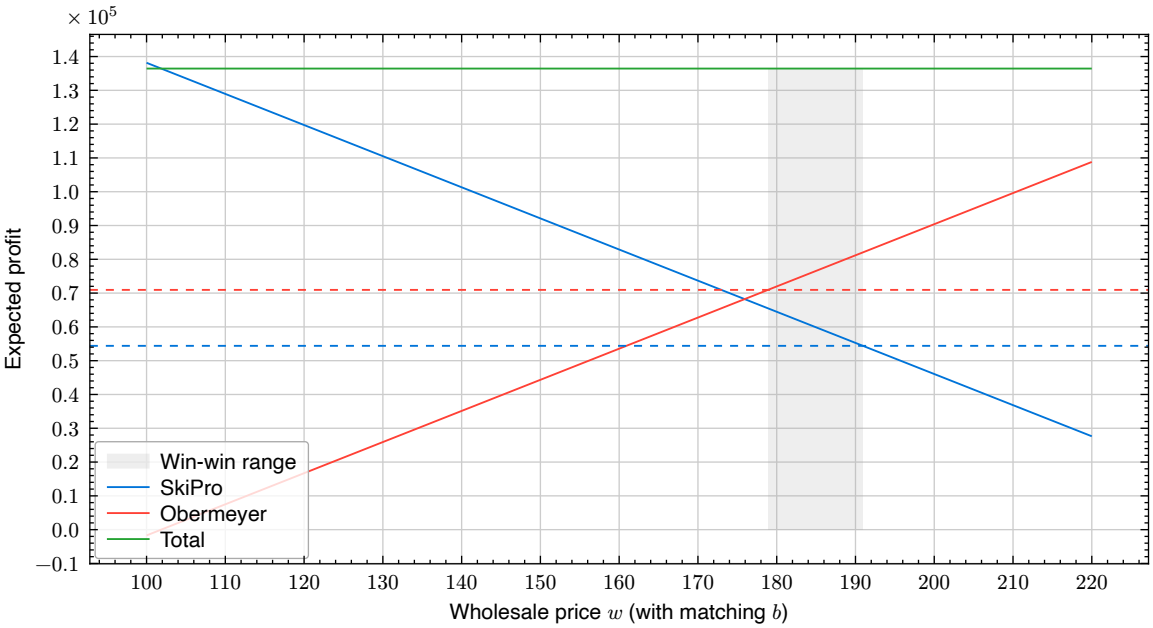
Raising w by 1 (with b from Step 2) moves m from SkiPro to Obermeyer:

$$m = Q^* - I\frac{p-s}{p-c} = 921$$

$$w_{\text{lo}} = w + \frac{\Pi_S^W - \Pi_S^{\text{BB}}}{m} = 178.9 \qquad w_{\text{hi}} = w + \frac{\Pi_R^{\text{BB}} - \Pi_R^W}{m} = 190.9$$

	w	b
Win-win range	178.9 – 190.9	172 – 186.1
Fair split (midpoint)	184.9	179.1

At the midpoint each firm gains 5550 over the wholesale contract.



1.4. SC Contracting

Double Marginalization Problem: Each firm makes decisions based on their own margin, not the supply chain's margin

Example: Sports Obermeyer

Wholesale contract

Obermeyer:
- Production cost (c): 100
- Revenue (w): 175

SkiPro
- Retail price (p): 250
- Slavage value (s): 75

Demand: $D \sim N(1000, 300)$

SKI PRO orders the optimal (profit maximizing) quantity:

$$c_u = 75$$
$$c_o = 100$$
$$\text{cr} = 0.4286$$
$$Q = 946$$

SKI PRO's Expected Profit:

$$\pi_R(Q; w) = pE[\text{Sold}] + sE[\text{Left}] - gE[\text{Short}] - wQ = 54392.14$$

Obermeyer's profit:

$$\pi_S(w; Q(w)) = (w - c)Q = 70949.72$$

Supply Chain expected profit:

$$\pi_{\text{SC}} = (w, Q) = \pi_R + \pi_S = 125341.86$$

Coordinated Supply Chain

Optimal quantity

$$c_u = p - c + g = 150$$
$$c_o = c - s = 25$$
$$\text{cr} = 0.86$$
$$Q^* = 1320.27$$

$$\pi_{\text{SC}}(Q; w) = pE[\text{Sold}] + sE[\text{Left}] - gE[\text{Short}] - cQ = 138153.63$$

Solution to Double Marginalization

- p: sell price
- w: wholesale price
- b: buyback price
- s: salvage value
- r: return cost

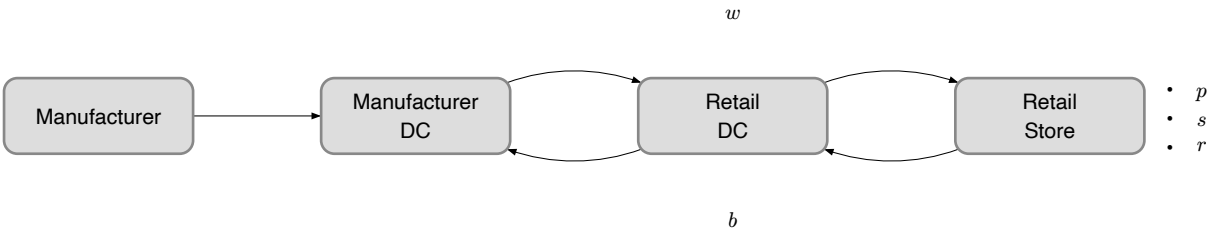
Share Risk / Buyback Contract

Obermeyer buys back unslod at $\$b$ / unit

SkiPro incurs cost r to ship jackets back

Conditions:
- $b < w$ so that returning doesn't make a profit
- $b > s + r$ so that return is preferable to salvage

Assume Obermeyer sets $w = 175$, what is b?



Obermeyer chooses b to make SkiPro's critical ratio equal to the integrated supply chain's critical ratio

$$\frac{p - w}{p - b + r} = \frac{p - c}{p - s} \quad \Rightarrow \quad b = p + r - (p - w)\frac{p - s}{p - c} = 167.5$$

	w	b
Feasible range	100 - 220	80 - 220
Win-win range	177 - 190.9	169.9 - 186.1
Fair split (midpoint)	184	178

Total gain from coordination: 12812 (6406 each at the midpoint).

$$\pi(Q) = \underbrace{(p-c)\mu}_{\text{Perfect Information}} \underbrace{- E[(p+g-c)(D-Q)^+ + (c-s)(Q-D)^+]}_{\text{Cost of Uncertainty (C(Q)}}$$

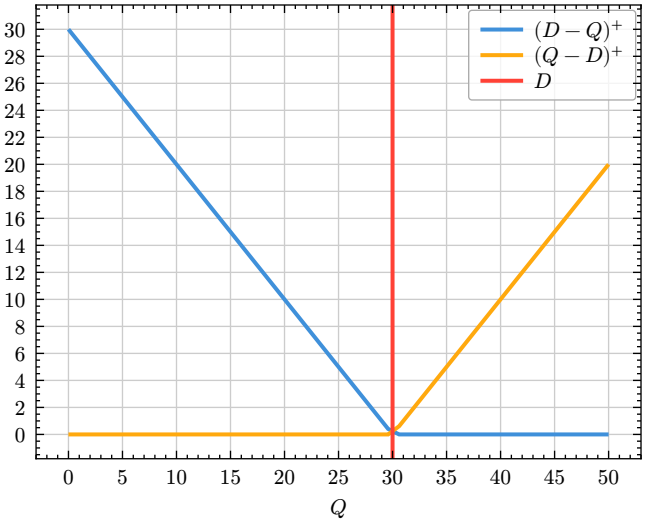
$$C(Q) = A(c_u, c_o) \cdot \sigma$$

c_u and c_o are given, only σ can be affected (increasing forecast accuracy)

$$\min C(Q) = E[c_u(D-Q)^+ + c_o(Q-D)^+]$$

$$\frac{\mathrm{d}C(Q)}{\mathrm{d}Q} = c_u(-1)\Pr(D > Q) + c_o(+1)\Pr(D \le Q) = 0$$

$$= -c_u(1 - \Pr(D \le Q)) + c_o \Pr(D \le Q) = 0 \quad \Rightarrow \quad \Pr(D \le Q^*) = \frac{c_u}{c_u + c_o}$$



Uniform Dist

$$\Pr(D \le Q^*) = 0.67 = \mathrm{cr}$$

$$F(Q^*) = \frac{Q^* - a}{b - a} = \mathrm{cr} \qquad D \sim U[0, 100]$$

$$Q^* = 67$$

Normal

$$P(D \le Q^*) = 0.4$$

$$z = \frac{Q - \mu}{\sigma} = -0.25$$

$$Q = \mu + z(\mathrm{cr}) \cdot \sigma = 94.93$$

$$L(z) = 0.54$$

$$E[\text{Short}] = \sigma L = 10.77$$
$$E[\text{Sales}] = \mu - E[\text{Short}] = 89.23$$
$$E[\text{Fill}] = \frac{E[\text{Sales}]}{\mu} = 0.8923$$
$$E[\text{Profit}] = pE[\text{Sold}] + sE[\text{Leftover}] - gE[\text{Short}] - cQ$$
$$= 86.38$$

If fill-rate target is given (F^b), what is the Q?

Poisson

x	$f(x)$	$F(x)$	$E[\text{Loss}]$

$$E[\text{Loss}] = \lambda(1 - F(Q-1)) - Q(1 - F(Q))$$

CDF:

- Look at y axis for critical ratios

- Look at corresponding x axis quantity
- That is the optimal Q

CR Plot:
- Qth unit ordered v. Expected Gain / Loss plot
- Expected Gain & Expected loss
- intersection is the CR

Critical Ratio
- c: cost per unit
- p: price per unit
- s: salvage price per unit leftover
- g: goodwill (penalty) per unit short
- c_u: Underage

$$c_u = p + g - c$$

- c_o: Overage

$$c_o = c - s$$

- cr: Critical Ratio

$$\mathrm{cr} = \frac{c_u}{c_u + c_o}$$

$$\Pi(Q) = E[p \cdot \min(D, Q) + s(Q - D)^+ - g(D - Q)^+] - cQ$$

Where:
- $(a)^+ = \max(0, a)$
- E: expectation because D is a random variable

$$\begin{aligned}\Pi(Q) &= E[p \cdot D - p(D - Q)^+ + s(Q - D)^+ - g(D - Q)^+ - cD + c(D - Q)^+ - c(Q - D)^+] \\ &= E[(p - c)D - \{(p + g - c)(D - Q)^+ + (c - s)(Q - D)^+\}] \\ &= (p - c)\mu - E[(p + g - c)(D - Q)^+ + (c - s)(Q - D)^+\}]\end{aligned}$$

$$\underbrace{(p-c)\mu}_{\substack{\text{profit with} \\ \text{perfect information}}} = \underbrace{E[\overbrace{(p+g-c)}^{c_u}(D-Q)^+ + \overbrace{(c-s)}^{c_o}(Q-D)^+]}_{\substack{\text{cost of} \\ \text{uncertainty}}}$$

$$\min C(Q) = E[c_u(D - Q)^+ + c_o(Q - D)^+]$$

$$\begin{aligned}\frac{\mathrm{d}C(Q)}{\mathrm{d}Q} &= c_u(-1)\Pr(D > Q) + c_o(+1)\Pr(D \le Q) \\ &= -c_u(1 - \Pr(D \le Q)) + c_o \Pr(D \le Q))\end{aligned}$$

Where:
- $\Pr(D \le Q) = F(x)$ (CDF)

$$-c_u(1 - \Pr(D \le Q)) + c_o \Pr(D \le Q)) = 0 \quad \Rightarrow \quad \Pr(D \le Q^*) = \frac{c_u}{c_u + c_o}$$

Plot:
- Production quantity v. Marginal loss/gain
- 2 curves:
  - Expected overage (marginal)
  - Expected underage (marginal)
- Their crossing is the optimal production quantity

$$\Pi(Q) = pE[\text{sold}] + sE[\text{leftover}] - gE[\text{short}] - cQ$$

Example: Oneil's Hammer 3/2 Economics

$$D \sim N(3192, 1181)$$

- p: 190
- c: 110
- s: 90
- c_u: $p - c = 80$
- c_o: $c - s = 20$

$$\text{cr} = \frac{c_u}{c_u + c_o}$$

$$\text{cr} = \Pr(D \leq Q^*) = 0.8$$

```
from scipy.stats import norm

mu = 3192
sigma = 1181

Q = norm.ppf(cr, loc=mu, scale=sigma)
```

Slide 13:

$$c_o \times F(Q) = c_u \times (1 - F(Q))$$

$$F(Q^*) = \Pr(D \leq Q^*) = \frac{c_u}{c_u + c_o} =$$

$$Q^* = \mu + z(\alpha) \cdot \sigma$$

Slide 17

1. In-stock Probability (Type I - service Level)

Probability all demand is satisfied

What is the in stock probability if the order is Q

$$z = \frac{Q - \mu}{\sigma}$$

$$\Pr(z)$$

We want 99% in-stock, what order quantity?

2. Expected Loss Sales (Shortage)

Slide 24

$$E[\text{short}] = E[(D - Q)^+] = \int$$

Normal Loss Function (Slide 25)

$$E[\text{short}] = \sigma \cdot L(z)$$

$$E[\text{sales}] = E[\min(D, Q)] = E[D - (D - Q)^+] = \mu - E[\text{short}]$$

$$E[\text{leftover}] = Q - E[\text{sales}] = Q - \mu + E[\text{short}]$$

$$\Pi(Q) = pE[\text{sold}] + sE[\text{leftover}] - gE[\text{short}] - cQ$$

3. Fill Rate (Type II - Service Level)

Fraction of demand that is fulfilled

$$\text{Fill Rate} = \frac{E[\text{sales}]}{\mu} = 1 - \frac{E[\text{short}]}{\mu}$$

Poisson

x	$P(D = x)$	CDF	$E[\text{Lost}]$

E[“Lost”] = E[“Short”] ?

- Demand random
- Single selling season
- Single opportunity to procure at the start of the season

1.5. Double Marginalization Problem

Each firm makes descision based on their own margin, not supply chain margin

Supplier:
- c: cost
- w: transfer price

Margin_s = w - c

Retailer:
- w: transfer price
- r: unit revenue

Margin_r = r - w

1.5.1. Case Study: Ski Jacket
- Cost c: 100
- Transfer Price w: 175
- Sell price p: 250
- Salvage value s: 75
- Goodwill g: 0
- Demand: $N \sim (1000, 300)$

$$c_u = p - w = 75$$
$$c_o = w - s = 100$$

$$\text{cr} = \frac{c_u}{c_u - c_o} = 0.4286$$

$$Q^* = F^{-1}(0.4286) = 946.02$$

SkiPro E[Profit]

$$\pi_{r(Q;w)} = pE[\text{Sold}] + sE[\text{Left}] + gE[\text{Short}] - cQ =$$

Obermeyer E[Profit]

$$\pi_s(w; Q(w)) = (w - c)Q =$$

Supply Chain E[Profit]

$$\pi_{\text{sc}} = \pi_r + \pi_s =$$

But if we integrate (coordinated SC):

$$c_u = p - c + g = 250 - 100 = 150$$
$$c_o = c - s = 100 - 75$$
$$0.4286 = \frac{150}{175} = 0.8571$$
$$Q = F^{-1}(0.8571) = 1320 \quad (39.5\% \text{ increase})$$
$$\pi_{\text{SC}} = 138154 \quad (10.22\% \text{ increase})$$

Pie chart (Integrated SC v. double marginalization impact)

1.5.1.1. Buyback contract r

$$b - r > s$$
$$b < w$$

$$c_u - p - w = 250 - 175 = 75$$
$$c_o = w - b + r = 180 - b$$

$$\frac{p - w}{p - b + r} = \frac{p - c}{p - s}(\text{integrated SC}) \quad \Rightarrow \quad b = p + r - (p - w)\frac{p - s}{p - c} = 167.5$$

1.5.1.2. Revenue Management with Capacity Controls
Matching supply to demand when supply is fixed

- Adjusting supply is not possible
- Customers can be segmented
- Number of …
- Q: Protection level
- $D < Q$: protect too many rooms (over protect) Rooms are empty which could have been sold to low fare travellers.
- $D > Q$: protect too few rooms (under protect). Some rooms could have been sold at the high fare instead of low fare.